

PROJECT OBJECTIVE

Build a pipeline that:

- ✓ Builds a .NET web app
- ✓ Uses **3 branches** (dev, main, release)
- ✓ Deploys sequentially (Dev → QA → Prod)
- ✓ Uses same artifact across environments

ARCHITECTURE

Developer → dev → Dev Env
→ main → QA Env
→ release → Prod Env

PROJECT SETUP

1. Create Repo

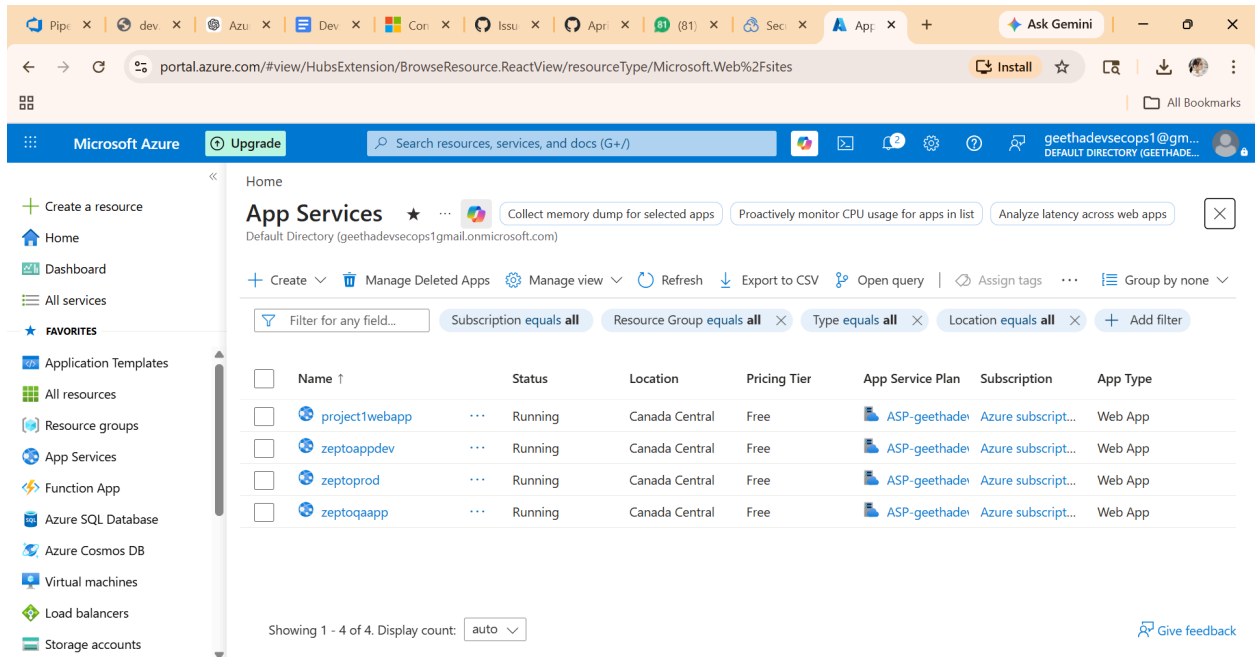
- Create project(zeptoecommerce) in **Azure DevOps**
- Add sample .NET app

2. Create Branches

- dev
- main (default branch)
- release

3. Create Azure Web Apps

- Dev App → **zeptoappdev**
- QA App → **zeptoqaapp**
- Prod App → **zeptoprodapp**



PIPELINE YAML (FINAL)

Open the vscode and make the changes in script as follows:

trigger:

```
branches:
  include:
    - dev
    - main
    - release
```

pool:

```
vmImage: 'windows-latest'
```

variables:

```
- name: solution
  value: '**/*.sln'
- name: buildPlatform
  value: 'Any CPU'
- name: buildConfiguration
  value: 'Release'
```

stages:

```
- stage: Build
  jobs:
    - job: BuildJob
      steps:
        - task: NuGetToolInstaller@1
        - task: NuGetCommand@2
```

```

    inputs:
      restoreSolution: '$(solution)'
- task: VSBUILD@1
  inputs:
    solution: '$(solution)'
    msbuildArgs: >
      /p:DeployOnBuild=true /p:WebPublishMethod=Package
/p:PackageAsSingleFile=true /p:SkipInvalidConfigurations=true
/p:DesktopBuildPackageLocation="$(Build.ArtifactStagingDirectory)\WebApp.zip" /p:DeployIisAppPath="Default Web Site"
    platform: '$(buildPlatform)'
    configuration: '$(buildConfiguration)'
- task: VSTEST@2
  inputs:
    platform: '$(buildPlatform)'
    configuration: '$(buildConfiguration)'
- task: CMDLINE@2
  displayName: "Check ZIP created"
  inputs:
    script: dir $(Build.ArtifactStagingDirectory)
- task: PUBLISHBUILDARTIFACTS@1
  inputs:
    PathToPublish: '$(Build.ArtifactStagingDirectory)'
    ArtifactName: 'drop'
    publishLocation: 'Container'
- stage: Deploy
  dependsOn:
    - Build
  condition: succeeded()
  jobs:
    - job: DeployJob
      steps:
        - task: DOWNLOADBUILDARTIFACTS@1
          inputs:
            buildType: 'current'
            downloadType: 'single'
            artifactName: 'drop'
            downloadPath: '$(System.ArtifactsDirectory)'
        - task: CMDLINE@2
          displayName: "Check downloaded ZIP"

```

```

    inputs:
      script: dir $(System.ArtifactsDirectory)\drop
- task: AzureRmWebAppDeployment@5
  inputs:
    ConnectionType: 'AzureRM'
    azureSubscription: 'Azure subscription
1(e99c6e01-0b8c-46e3-be07-1dcaf7ff891c) '
    appType: 'webApp'
    WebAppName: 'zeptoappdev'
    package: '$(System.ArtifactsDirectory)/drop/WebApp.zip'
- stage: QA
  dependsOn:
    - Build
  condition: and(succeeded(),or(eq(variables['Build.SourceBranch'],
'refs/heads/main'),eq(variables['Build.SourceBranch'],
'refs/heads/release'))))
  jobs:
    - job: QAJob
      steps:
        - task: DownloadBuildArtifacts@1
          inputs:
            buildType: 'current'
            downloadType: 'single'
            artifactName: 'drop'
            downloadPath: '$(System.ArtifactsDirectory)'
        - task: AzureRmWebAppDeployment@5
          inputs:
            ConnectionType: 'AzureRM'
            azureSubscription: 'Azure subscription
1(e99c6e01-0b8c-46e3-be07-1dcaf7ff891c) '
            appType: 'webApp'
            WebAppName: 'zeptoqaapp'
            packageForLinux: '$(System.ArtifactsDirectory)/drop/*.zip'
- stage: Prod
  dependsOn:
    - Build
  condition: and(succeeded(), eq(variables['Build.SourceBranch'],
'refs/heads/release'))
  jobs:
    - job: ProdJob

```

```
steps:
- task: DownloadBuildArtifacts@1
  inputs:
    buildType: 'current'
    downloadType: 'single'
    artifactName: 'drop'
    downloadPath: '$(System.ArtifactsDirectory) '
- task: AzureRmWebAppDeployment@5
  inputs:
    ConnectionType: 'AzureRM'
    azureSubscription: 'Azure subscription
1(e99c6e01-0b8c-46e3-be07-1dcac7ff891c) '
    appType: 'webApp'
    WebAppName: 'zeptoprod'
    packageForLinux: '$(System.ArtifactsDirectory)/drop/*.zip'
```

Dev Branch

Build → Dev

Main Branch

Build → Dev → QA

Release Branch

Build → Dev → QA → Prod

Run pipeline:

Now commit and push the changes to Azure devops and run the pipeline. Once the pipeline is successfully run , then validate whether the project is successfully deployed on 3 webAPPS.

Pipeline: x dev.azure x Home P x Azure P x Git - Ins x Devsec x (79) Wh x zepto x + Ask Gemini x

dev.azure.com/geethadevsecops1/zeptoecommerce/_build/results?buildId=71&view=results

Azure DevOps geethadevsecops1 / zeptoecommerce / Pipelines / zeptoecommerce (9) / 20260502.14

zeptoecommerce

- Overview
- Boards
- Repos
- Pipelines
- Pipelines
- Environments
- Releases
- Library
- Task groups
- Project settings

Summary Associated pipelines Code Coverage

Stages Jobs

Build 1 job completed 1m 39s 1 artifact

Deploy 1 job completed 47s

QA 1 job completed 36s

Prod 1 job completed 45s

Dev:

Home Page - T x Azure Pipelines x Git - Install for x Devsecops-assi x (79) WhatsApp x zeptoqaapp - x + Ask Gemini x

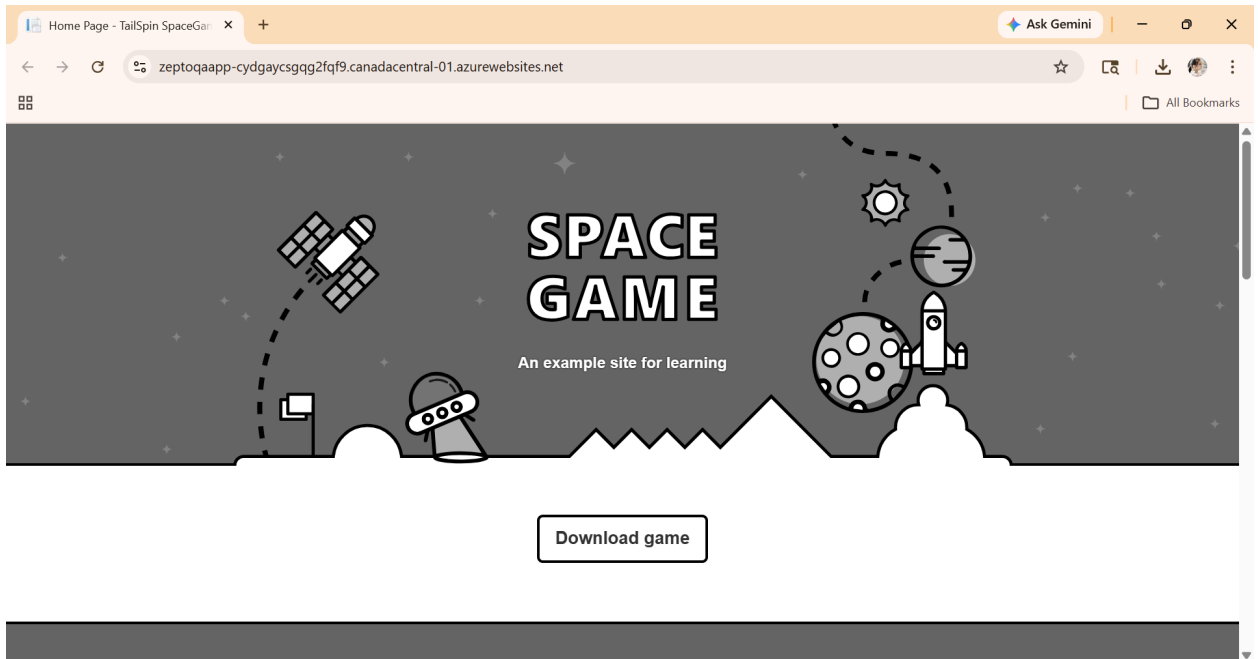
zeptoappdev-hghffefkfwfxe4hs.canadacentral-01.azurewebsites.net

SPACE GAME

An example site for learning

Download game

QA:



Prod:

